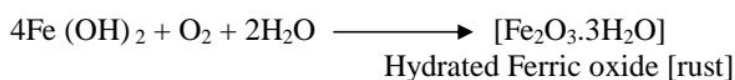


Module-2**Corrosion and Metal Finishing****Corrosion****Definition of corrosion:-**

Corrosion can be defined as “The destruction (or) deterioration (or) loss of metals or alloys by the surrounding environment through chemical (or) electrochemical reactions”.

Ex: - 1. Rusting of Iron: - A reddish brown scale formation on iron objects.



2. Green scales formed on copper vessels.

Corrosion may occur either in a dry environment (or) in an aqueous medium. The former is called dry corrosion and the later is called wet corrosion.

Dry corrosion [chemical corrosion]

Dry corrosion involves the direct attack of metals by dry gases mainly through chemical reactions.

It occurs by direct attacks of atmospheric gasses such as oxygen, hydrogen, sulphide, halogens and Sulphur-dioxide on the metal forming oxide layer.

Ex: - the attack of dry air (or) oxygen on a metal to form an oxide layer over the surface.

Wet corrosion [Electrochemical corrosion]

Wet corrosion involves reactions in aqueous solution medium. The conducting surface of the metal undergoes an electrochemical reaction with the moisture and oxygen present in the atmosphere.

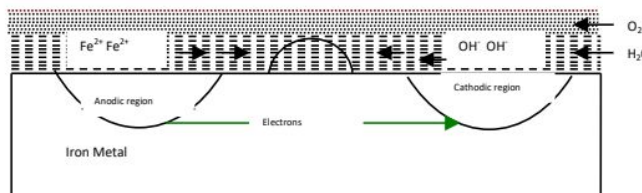
Ex: - Rusting of Iron.

Wet corrosion is explained on the basis of electrochemical theory.

Electrochemical theory of corrosion:-

Electrochemical Theory taking Iron as example:-

According to electrochemical theory, when a metal such as iron is exposed to corrosive environment, following changes occur. A large number of tiny galvanic cells with anodic and cathodic regions are formed.

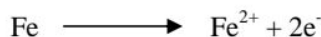


At the anodic region oxidation reaction takes place and the metal gets converted into its ions, liberating electrons. Consequently, metal undergoes corrosion at the anodic region.

At the cathodic region, reduction reaction takes place. Since the metal cannot be reduced further, metal atoms at the cathodic region are unaffected by the cathodic reaction.

At Anode:-

At anode oxidation takes place in which the metal atoms are converted into their ions liberating electrons.



At Cathode:-

The cathodic reaction is based on nature of the environment.

a). In the absence of oxygen.[Libration of Hydrogen]

i) In acidic medium, liberation of hydrogen takes place.

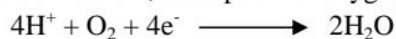


ii) In neutral (or) alkaline medium, Hydroxide ions are formed with simultaneous liberation of hydrogen.

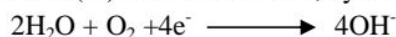


b). In presence of oxygen:- [Absorption of oxygen]

i). In acidic medium, absorption of oxygen takes place.



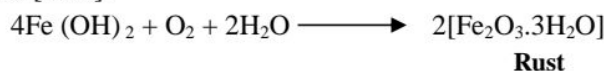
ii). In neutral (or) alkaline medium, hydroxide ions are formed.



Then Iron ions (Fe^{2+}) from anode combine with hydroxyl ions from cathode to form iron hydroxide on the surface, b/w the anode and cathode areas.



Iron hydroxide further reacts with oxygen and water forming hydrated iron oxide which is the corrosion product [Rust].



Types of Corrosion:-

Differential Metal Corrosion:-

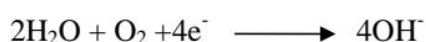
Differential metal corrosion occurs when two dissimilar metals are in contact with each other and exposed to a corrosive conducting medium, the metal with lower electrode potential acts as anode undergoes oxidation and gets corroded. and the other metal with higher electrode potential acts as cathode. Undergo reduction & protected from corrosion. This type of corrosion is known as galvanic corrosion (or) differential metal corrosion.

The rate of corrosion depends mainly on the difference in potential b/w two metals. Higher the difference faster is the rate of corrosion.

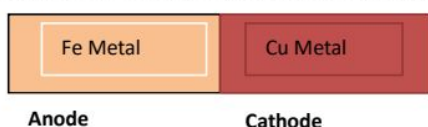
The reactions may be represented as,

At anode: $- M \longrightarrow M^{n+} + e^{-}$

At cathode:- reduction reaction takes place, depending upon the nature of the environment.



Ex:-i). When iron is in contact with copper [0.34V], iron [-0.44V] becomes anodic and undergoes corrosion whereas copper becomes cathodic and remains unaffected.



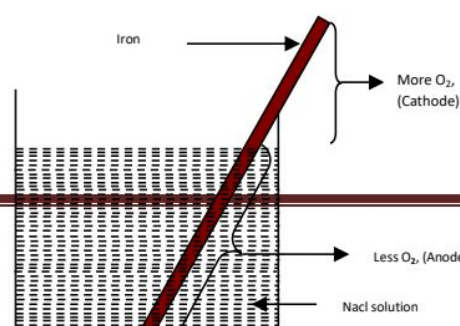
- ii). Tin coating copper vessel.
- iii). Steel pipe connected to copper plumbing.
- iv). Bolt and nut made of different metals.

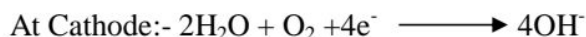
Differential Aeration Corrosion:-

When a metal is exposed to different concentrations of air (or) oxygen, part of the metal exposed to lower concentration of oxygen becomes anodic and undergoes corrosion. Whereas, other part of the metal exposed to higher concentration of oxygen becomes cathodic and remains unaffected. This kind of corrosion is called as differential aeration corrosion.

Ex:- Iron rod partially immersed in NaCl solution, part of the metal immersed in solution is exposed to lower concentration of oxygen becomes anodic and undergoes corrosion. Whereas part of the metal outside NaCl solution, is exposed to more oxygen becomes cathodic and remains unaffected.

The anodic and cathodic reactions are





Other Examples:-

1. Part of the nail inside the wall undergoes corrosion.
2. When a dust particle sits on a metal bar, the part under the dust undergoes corrosion.
3. Partially filled iron tank undergoes corrosion inside water.

Typical examples of differential aeration corrosion are the Water-line corrosion & Pitting corrosion.

a). Water-line Corrosion:-

It is observed in steel (or) Iron water tanks partially filled with water.

When a steel tank is partially filled with water for a long time, the inner portion of the tank below the water line is exposed only to dissolve oxygen, whereas the portion above the water line is exposed to more oxygen. Thus the portion below the water line acts as anode and undergoes corrosion. The upper portion acts as cathode and is unaffected.

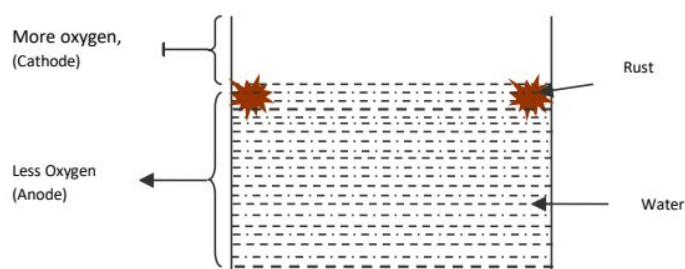
A distinct brown line is formed just below the water line due to the deposition of rust.

The cell reactions are



Finally, the ions combine to form corrosion product.

Ex: - Ships which remain partially immersed in sea water for a long time undergo water line corrosion.



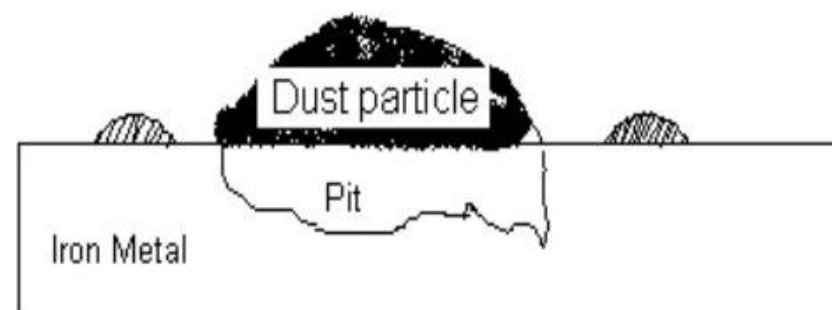
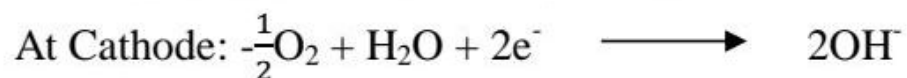
b). Pitting Corrosion:-

This type of corrosion is observed when dust particles (or) oil drops deposited over the metal surface.

Pitting corrosion results when the portion of the metal covered by dust which is less aerated becomes anodic and undergoes corrosion and form pit. The remaining area of the metal which is exposed to higher concentration of oxygen becomes cathodic and remains unaffected.

Once a small pit is formed the rate of corrosion increases because formation of small anodic area [pit] to corrode faster because accepting electrons from the cathodic area increases.

The cell reactions are



Note:- higher the conductivity of the medium, faster the ion can migrate between the anodic and Cathodic regions of the cell.

Corrosion Control Methods:-

Corrosion can be controlled by preventing the formation of galvanic cells on the surface of metals.

The methods to control corrosion can be classified as,

1. Design & Selection of materials.
2. Protective coatings. i). Inorganic coatings. ii). Metal coatings.
3. Cathodic Protection.

1. Protecting Coating:-

Corrosion is prevented by the application of protective coating on the surface of metal, thereby the metal surface is isolated from the corrosive environment. Important types of protective coating are,

- a. Metal coating.
- b. Inorganic coating.

a.Metal coating:-

“The process of coating base metal with a layer of protective metal is known as metal coating”.

There are two types of metal coating,

1. Anodic coating.
2. Cathodic coating.

1.Anodic coating:-

“The coating metal is anodic to base metal is called anodic coating”.

Ex: - Galvanization.

Galvanization:-

“The process of coating zinc on the base metal [Fe, Steel] surface by hot dipping is known as galvanization”.

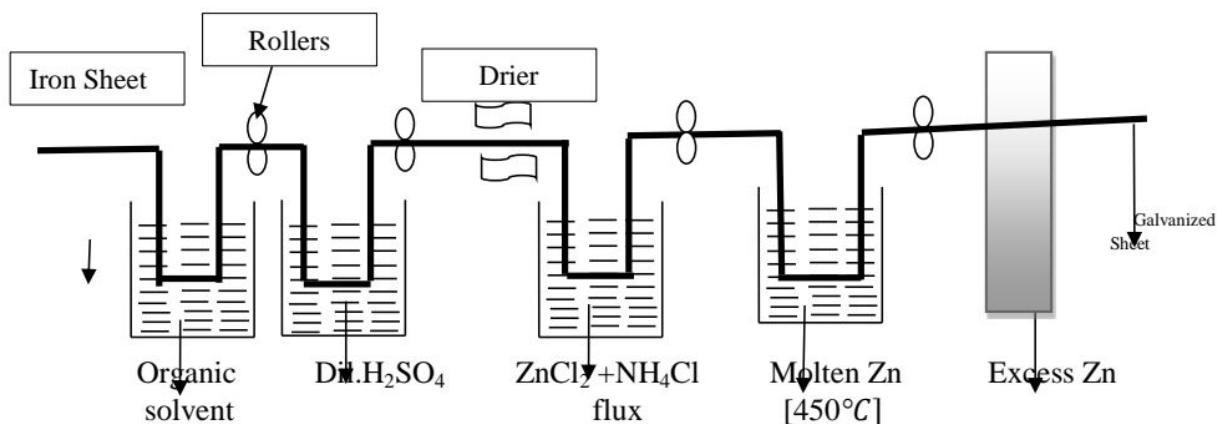
The galvanization process is carried out as follows

- The metal surface is washed with organic solvent[CCl₄, Toulene,Benzene] to remove organic matters.
- The metal surface is treated with dil.H₂SO₄ to remove rust and other deposits.
- Then it is well washed with water & it is air dried by passing with hot air.
- The metal is treated with a mixture of zinc chloride and ammonium chloride solution which prevents the oxidation of the coated metal.
- Then the metal is dipped in molten Zinc at 450⁰c
- Excess of zinc on surface is removed by passing a pair of hot rollers, which removes excess of zinc and produced thin coating

Iron Sheet

Rollers

Drier



Application: - Galvanized materials are used in fencing wire, buckets, bolts, nuts, nails, screw etc.

Note:- Galvanized sheets cannot be used for preparing (or) storing food because zinc dissolves in acidic medium and forms toxic compounds.

b. Inorganic coating:-

These coatings are produced at the surface of the metal by chemical (or) electrochemical reactions. **Ex:** - Anodizing and Phosphating.

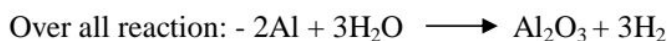
Anodizing:-

Anodizing coatings are generally produced on non-ferrous metals like Al, Zn, Mg and their alloys by anodic oxidation processes in which the base metal is made as anode.

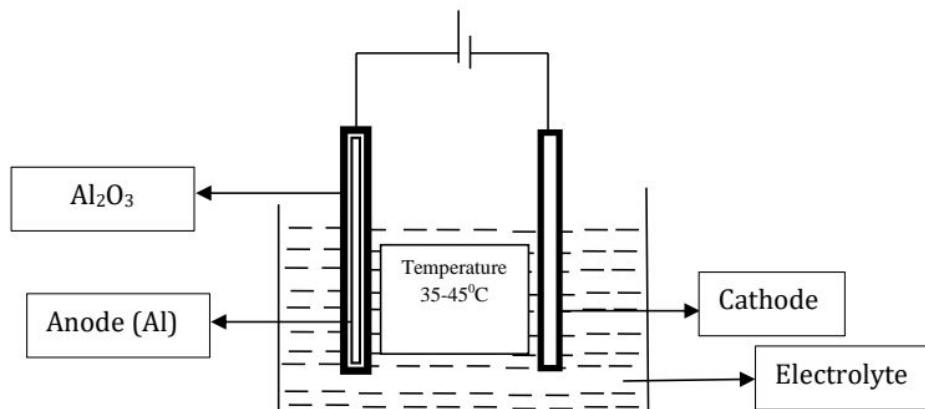
Anodizing of Aluminum:-

When aluminum metal is made anodic in an electrolytic bath with H_2SO_4 (or) chromic acid as the electrolyte, a thin layer of Al_2O_3 is formed on the surface. This process is called Anodizing (or) Anodic oxidation of Aluminum.

The porous coating is obtained by anodic oxidation; it is carried out by making it anode in an electrolytic bath containing a suitable acid like chromic acid (or) H_2SO_4 at 35°C - 40°C . A plate of lead (or) stainless steel is made the cathode. When current of moderate density is passed, the O_2 liberated at the anode combines with it to form oxide which takes the form of thick film Al_2O_3 deposits on the surface of the object.



The coating is slightly porous and it is sealed by dipping in hot water. Al_2O_3 gets hydrated to $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$ which occupies more volume and hence protects the metal from corrosion.



Application: - Metals such as Al, Mg, Ti etc are anodized to control corrosion.

Cathodic Protection:-

Cathodic protection is the technique of offering protection to a specimen against corrosion by providing electrons from an external source.

There are two important methods of cathodic protection.

1. Sacrificial Anode Method.
2. Impressed Current /Voltage Method.

1. Sacrificial Anode Method:-

Sacrificial anode method involves the use of a metal which is anodic to the specimen.

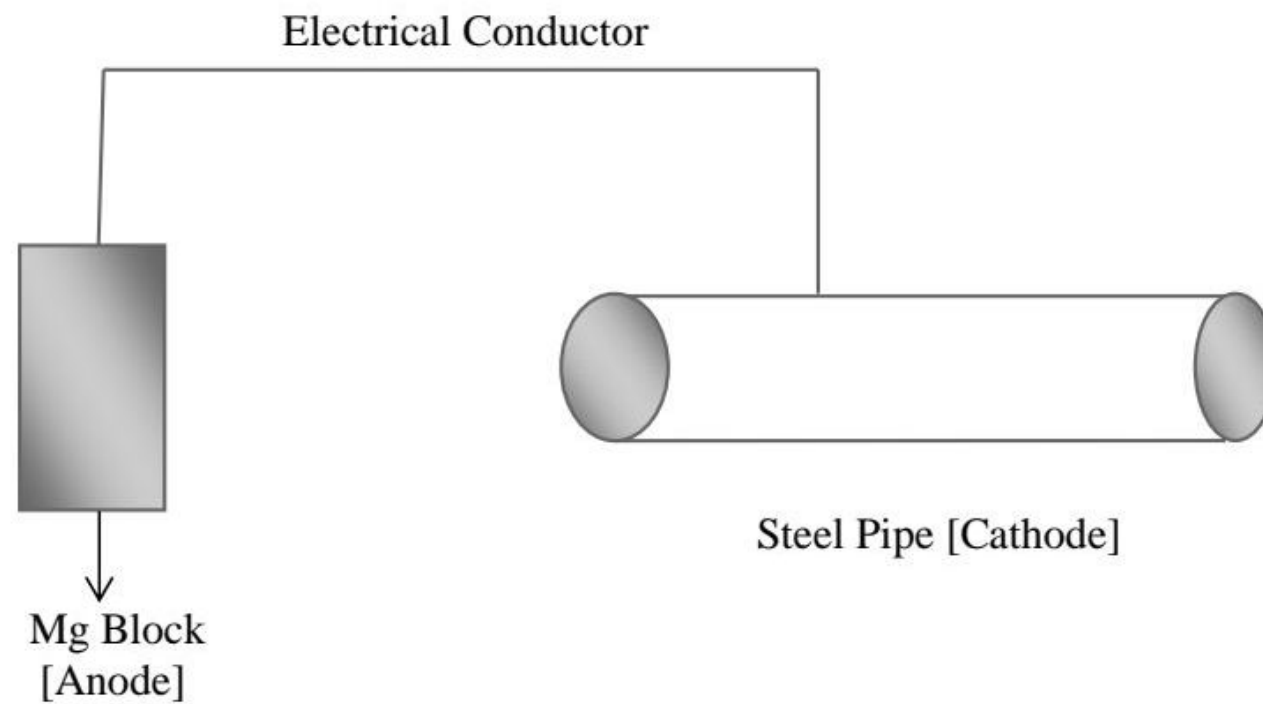
In this method the metallic structure to be protected is connected to a more anodic metal using a metallic wire. More active metal gets corroded, while the parent metallic structure is protected from corrosion.

Ex: - when steel metal to be protected, it may be connected to a block of Mg or Zn, in such case, steel acts as cathode and is unaffected. Mg or Zn act as anode and undergo sacrificial corrosion. When the sacrificial anode gets exhausted, it is replaced with new ones.

The commonly used sacrificial anodes are Mg, Al, Zn, and their alloys.

Ex: - 1. Protection of ship's hull by fixing Zinc plates.

2. Protection of an underground pipeline with a magnesium anode.



EMF of the Cell: The potential difference between the two electrodes of a galvanic cell which causes the flow of current/ electrons from one electrode to the other is called the emf of the cell. It is denoted by E_{cell} .

Standard EMF of the Cell: E°_{cell} is defined as the emf of a galvanic cell when the reactants and products of the cell reactions are at unit concentration or unit activity at 298 K and at 1 atmospheric pressure.

Free Energy

A part of the total energy of a system is converted into work and the rest is unavailable. So, any kind of energy which can be converted into useful work is called 'available energy'.

Single Electrode Potential

The potential difference between the metal and the solution at the interface is called the single electrode potential.

REFERENCE ELECTRODES

Definition:

A reference electrode is an electrode which has a stable and known electrode potential with reference to which the electrode potential of any other electrode can be determined.

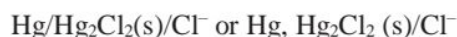
There are two types of reference electrodes.

1. Primary Reference Electrode (SHE)
2. Secondary Reference Electrodes (Ag-AgCl and Calomel)

CONSTRUCTION AND WORKING OF CALOMEL ELECTRODE

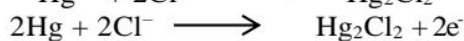
Calomel electrode is a secondary reference electrode. It belongs to the category of metal-metal insoluble salt electrode. Calomel is the trade name of Hg_2Cl_2 and it is sparingly soluble in water. Calomel electrode consists of a glass tube in which thin layer of pure mercury is placed at the bottom. It is covered by a paste of mercury, mercurous chloride with KCl. A platinum wire sealed into a glass tube is dipped into mercury and used to provide the external electrical contact. A solution of KCl is introduced above the paste through the side tube shown on the right. The concentration of KCl solution used is either decinormal, normal or saturated. The solution also fills in the side tube on the left. This constitutes calomel electrode. It is connected to the other half cell through the side tube having the jet or through a salt bridge.

Calomel electrode is represented as



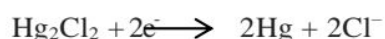
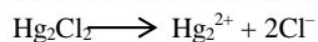
Calomel electrode is a reversible electrode. Depending on the nature of the other electrode with which it is connected, calomel electrode can act as anode or cathode.

Electrode Reactions



As cathode

The electrode reaction is reverse of the above

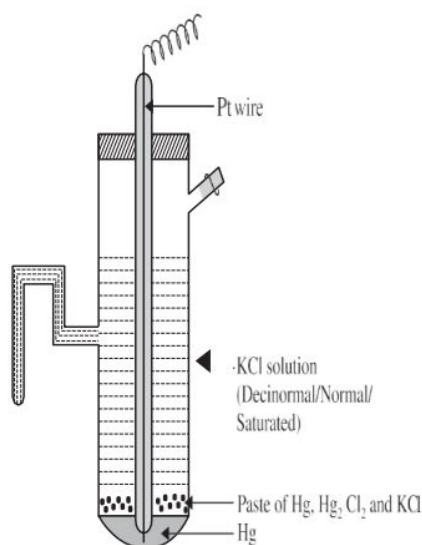


The net reversible electrode reaction is $\text{Hg}_2\text{Cl}_2 + 2\text{e}^- \leftrightarrow 2\text{Hg} + 2\text{Cl}^-$

Nernst equation for calomel electrode is

$$E = E^\circ + \frac{2.303RT}{nF} \log [\text{Cl}^-]$$

The potential of calomel electrode is dependent on the concentration of chloride ions and the electrode potential is reversible with the concentration of chloride ions.



Concentration of KCl solution	Name	EMF of calomel electrode at 298 K
0.1 N	Decinormal Calomel Electrode (DNCE)	0.3334 V
1 N	Normal Calomel Electrode (NCE)	0.281V
Saturated	Saturated Calomel Electrode (SCE)	0.2422V

Advantages

1. It is simple to construct.
2. The electrode potential is constant and reproducible.
3. The potential does not vary much with temperature.

Applications

1. It is used as a secondary reference electrode in labs in the measurement of single electrode potential.
2. It is the most commonly used reference electrode in all potentiometric determination.
3. It is used in corrosion studies.

ION SELECTIVE ELECTRODES

Ion selective electrodes can selectively detect and quantitatively measure a particular ion in a mixture neglecting other ions and the potential developed is a function of the concentration of that particular ion. These electrodes usually consist of a membrane which is capable of exchanging a specific ion with which it is in contact. Therefore, these electrodes are also known as membrane electrodes.

GLASS ELECTRODE

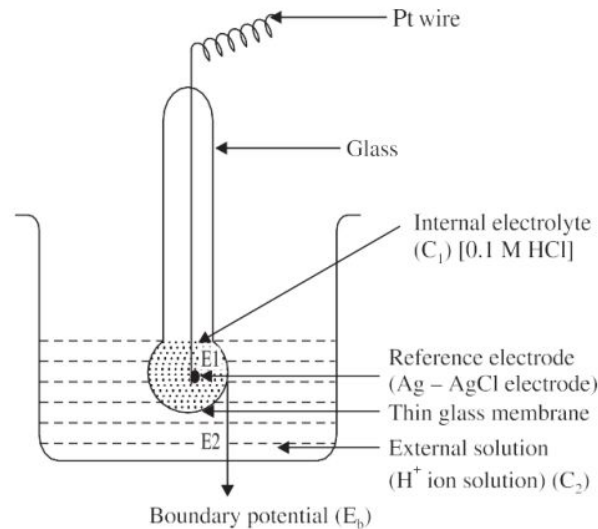
It is an example of ion selective electrode. This electrode is responsive towards H^+ and extensively used to measure the pH of the solution. It is the oldest and the most widely used electrode for pH determination.

Principle

When a thin glass bulb containing an acid of one concentration is immersed in a solution of an acid with different concentration, a potential across the glass membrane is developed.

Construction

It consists of a long glass tube with a thin-walled bulb at one end as shown in the diagram. The composition of the glass used in glass electrode is $\text{Na}_2\text{O} - 22\%$, $\text{CaO} - 6\%$ and $\text{SiO}_2 - 72\%$. It has a low melting point, high electrical conductance and can sense hydrogen ions upto a pH of about 9. The glass bulb contains usually 0.1M HCl as internal standard solution. Ag/AgCl electrode (as internal reference electrode) is dipped in the solution and connected by a wire of Pt for electrical contact.



The bulb of the glass electrode is dipped in either distilled water or dilute HCl. The glass surface should be always hydrated.

The mobility of H^+ through the hydrate layer is rapid as compared with the dry layer. This helps in the quick attainment of equilibrium.

The glass electrode can be represented as:



When a thin walled glass bulb containing an acid is immersed in another solution containing H^+ ions, a potential is developed across the glass membrane. This is known as the boundary potential (E_b). It is due to the difference in the concentration of H^+ ions in two solutions (i.e., C_1 and C_2 where $C_1 = 0.1 \text{ M}$).

Boundary potential is given by the relation.

$$E_b = E_2 + E_1 = \frac{RT}{F} (\ln C_2 - \ln C_1)$$

Where C_1 and C_2 are the concentration of the H^+ of the solution inside and outside the glass bulb respectively.

$$E_b = \frac{RT}{F} \ln C_2 - \frac{RT}{F} \ln C_1$$

The concentration of C_1 is constant.

Therefore $E_b = \text{constant} + \frac{RT}{F} \ln C_2$

Substituting the values of R and F and at 298 K and converting natural logarithm to base 10

$$E_b = L + 0.0591 \log C_2 \quad \dots(1)$$

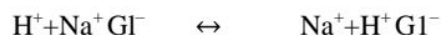
Where L is a constant which depends on the pH of the solution taken in the bulb and the glass electrode assembly. Since $C_2 = [\text{H}^+]$ of the solution and $-\log [\text{H}^+] = \text{pH}$

Equation (1) becomes

$$E_b = L - 0.0591 \text{pH}$$

Working of Glass Electrode

When the glass electrode is dipped in the test solution, Na^+ ions of the glass membrane are exchanged for H^+ of the test solution.



(Solution)(Membrane)(Solution)(Membrane)

The boundary potential developed due to the above exchange is mainly responsible for the glass electrode potential E_G

$$\text{WKT } E_b = E_2 - E_1 = \frac{RT}{F} (\ln C_2 - \ln C_1)$$

$$E_G = E_b + E_{\text{Ag/AgCl}}$$

When $C_1 = C_2$

$E_b = 0$

But it has been observed that even when $C_1 = C_2$, a small potential is developed. This is called asymmetric potential. Hence, potential of glass electrode is the sum of three potentials

Boundary potential (E_b)

Potential of Ag/AgCl electrode ($E_{\text{Ag/AgCl}}$)

Asymmetric potential (E_{asy})

$$E_G = E_b + E_{\text{Ag/AgCl}} + E_{\text{asy}}$$

$$E_G = L - 0.0591 \text{ pH} + E_{\text{Ag/AgCl}}$$

$$E_G = L_1 - 0.0591 \text{ pH} \quad \dots(4)$$

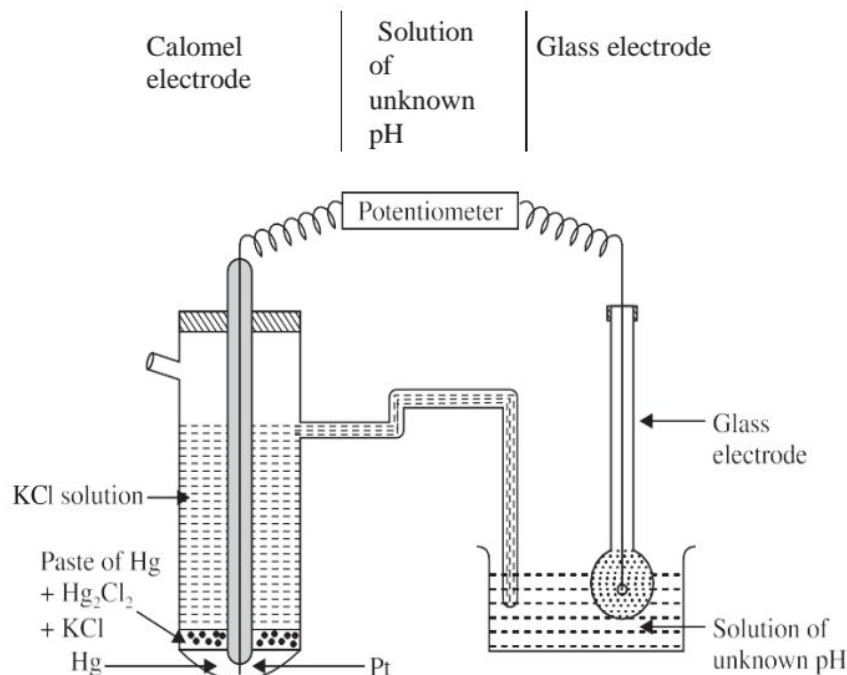
Where L_1 is a constant and equal to $L + E_{\text{Ag/AgCl}} + E_{\text{asy}}$

DETERMINATION OF pH USING GLASS ELECTRODE

In order to measure the pH of a given solution, a galvanic cell is constructed by combining glass electrode with saturated calomel electrode (SCE). The cell is represented as

H g (l)	Hg ₂ Cl ₂ (s)	KCl	Solution of unknown pH	Glass membrane	HCl (0.1M)	AgCl	A g (S)
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or



The potential of the above cell (E_{cell}) is measured using a potentiometer circuit or a pH meter.

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$E_{\text{cell}} = E_{\text{G}} - E_{\text{SCE}}$$

(By convention, the glass electrode assembly is treated as cathode) From equation (4)

$$E_{\text{G}} = L_1 - 0.0591 \text{ pH}$$

$$E_{\text{cell}} = L_1 - 0.0591 \text{ pH} - 0.2422$$

$$\text{pH} = \frac{L_1 - 0.2422 - E_{\text{cell}}}{0.0591}$$

Advantages of Glass Electrode

Glass electrode is a versatile analytical tool and has distinct advantages.

1. It is a stable electrode system and can be used in the presence of strong oxidising or reducing substances.
2. It is portable and compact.
3. It is not poisoned easily.
4. It is simple to operate.
5. It can be used in the presence of viscous biological fluids.
6. It can be used in portable instruments.
7. It can detect and estimate H^+ in presence of other types of ions. Accurate results are obtained between pH range 1 and 9. However by using special glass electrode pH 1 to 14 can be measured.

Disadvantages of Glass Electrode

1. The electrode can be used up to pH 13 but becomes sensitive to Na^+ above pH 9 resulting in alkaline error.
2. The glass membrane is very thin, so it must be handled with utmost care.
3. Fluoride ions in the test solution attack the glass surface and alter the composition of the membrane.
4. It does not function satisfactorily in pure alcohols.
5. Due to high resistance of glass, simple potentiometers cannot be used. It requires sensitive electronic potentiometers for EMF measurements.
6. In acidic solution of $\text{pH} < 1$, the salt effects due to anions are observed.

Potentiometry

The measurement of EMF to determine the concentration of ionic species in solution is referred to as potentiometry.

Potentiometry is used to determine the concentration of a solution.

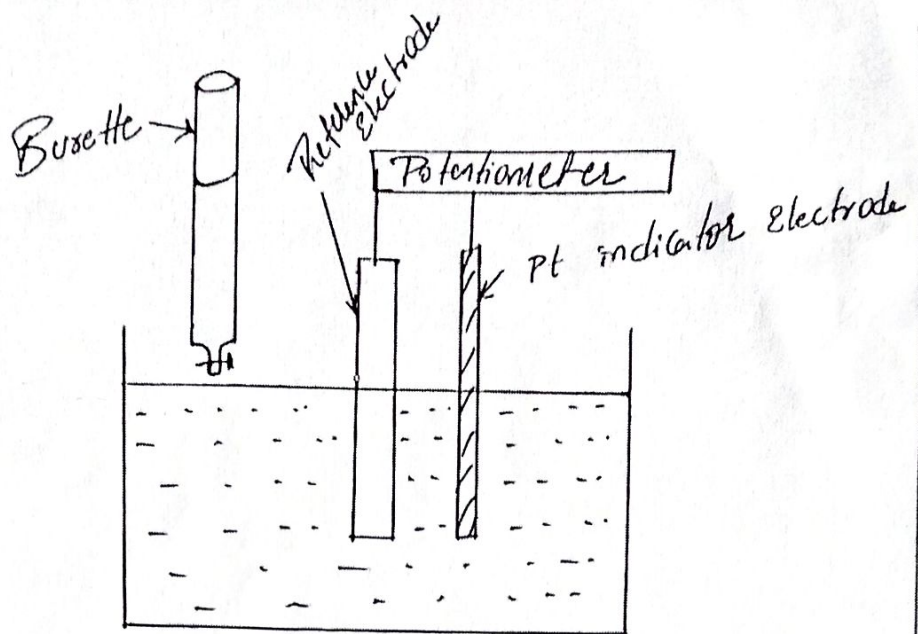
Theory :- * The relation between electrode potential & metal ion concentration is given by the Nernst Equation

$$E = E^{\circ} + \frac{0.0591}{n} \log [M^{n+}]$$

- * EMF can be measured by combining the electrode with reference electrode.
- * The EMF of the cell changes gradually till the end point & changes rapidly at very close to the end point.
- * The equivalence point can be determined graphically by plotting change in potential against volume of titrant added.
- * From the equivalence point calculate concentration & amount of substance in solution.

Instrumentation :-

- * Potentiometer consists of a reference electrode, an indicator electrode & a device for measuring the potential.
- * A known volume of the analyte is taken in a beaker & its potential is determined by connecting the assembly to a potentiometer.
- * The titrant is added in increments of 0.5 ml & the potential is measured each time.
- * The end point is determined by plotting change in potential against the volume of titrant.

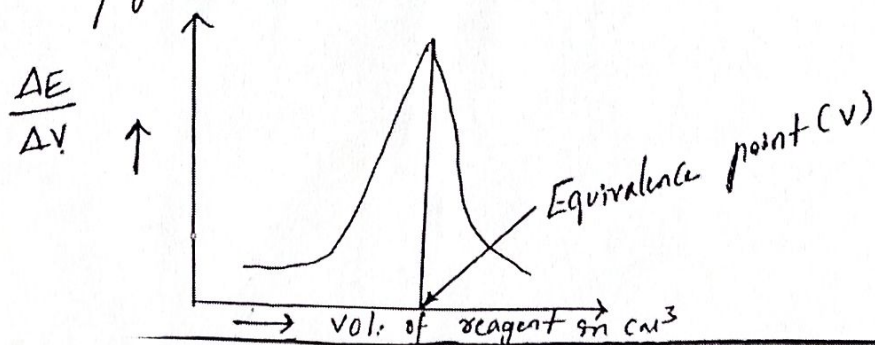


Applications :-

It finds applications in oxidation reduction titrations to estimate the concentration of analyte in the sample solution.

Example :- Estimation of amount of FAS by using Potentiometer method.

- * The FAS solution is taken in the beaker & $K_2Cr_2O_7$ is taken in the micro burette.
- * Immerse the electrodes in the beaker along with stirrer.
- * The electrodes are connected to the potentiometer which gives the emf values.
- * Go on adding the titrant in the interval of 0.5 ml.
- * There will be sudden jump in the emf values after adding some amount of reagent, take five more readings.
- * Plot a Graph $\Delta E / \Delta V$ v/s vol. of reagent added.



Conductometry

(8)

* Conductometry is an electrochemical method of analysis based on measuring the resistance of an electrolyte sol.

* Reciprocal of resistance is called conductance i.e. $C = 1/R$
Unit is 'Siemens' (or) ohm^{-1}

Specific conductance (κ) It is defined as "the conductance of the solution present b/w two parallel electrodes of 1 cm^2 area of cross section & 1 cm apart". $R = \rho \frac{l}{a}$

Theory :-

* Conductance measurements are used to determine the Equivalence point in acid-base titrations.

* In which there is sudden change in conductance of the solution after the Equivalence point.

* The Equivalence point is determined graphically by plotting conductance against the titre values.

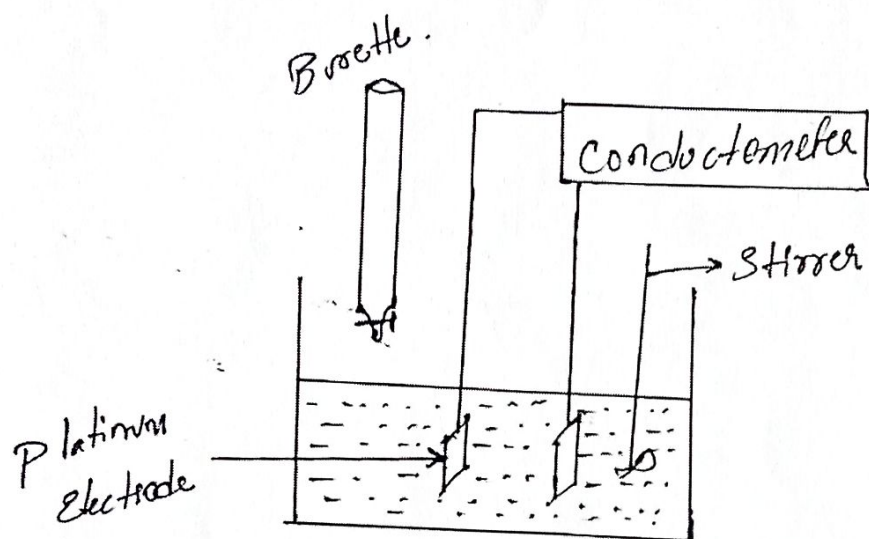
* In conductometric titrations ions of a particular conductance are replaced by ions of different conductance.

Instrumentation :-

* It consists of two Pt-electrodes each of unit area of cross section placed unit distance apart.

* The electrodes are dipped in the electrolyte solution taken in a beaker.

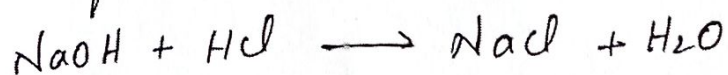
- * It is connected to a conductance measuring device.
- * The titrant is added from a burette & the solution is stirred.
- * The conductance is measured after the addition of the titrant at intervals of 0.5 mL (or) 1 mL.



Applications :-

① Strong acid with strong base (HCl v/s NaOH)

- * Consider the titration of strong acid HCl with strong base NaOH.

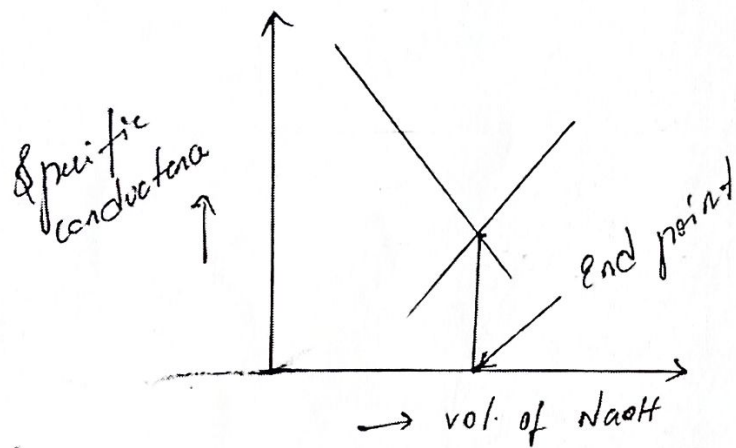


- * On addition of NaOH, the conductance of the acid solution decreases gradually until the acid is completely neutralized.

* The Conductance decreases, because the small & highly mobile H^+ in the acid is gradually replaced by heavier & less mobile Na^+ ions.

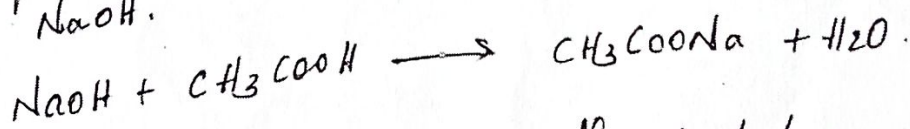
* Addition of base beyond the neutralization point leads to increase in the number of more mobile OH^- ions & Conductance increases.

* A plot of conductance against the volume of $NaOH$ is shown in the figure. The point of intersection of the two lines is the neutralization point.



② Weak acid with strong base. (CH_3COOH v/s $NaOH$)

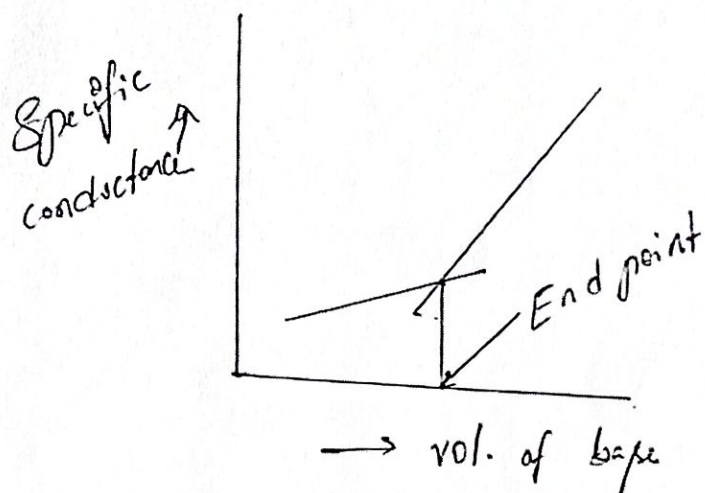
* Consider the titration of acetic acid with a strong base $NaOH$.



* On addition of $NaOH$, the conductance of the weak acid solution increases until the acid is completely neutralized.

* The conductance increases, because the partially dissociated, weakly conducting weak acid gradually becomes fully dissociating sodium acetate.

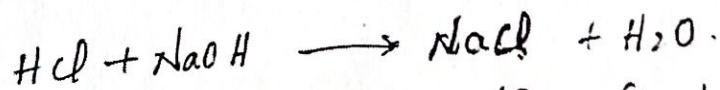
- * Addition of base beyond the neutralization point leads to increase in the number of highly mobile OH^- ions. Hence the conductance increases sharply.



③ Acid mixture with a strong base (HCl & CH_3COOH v/s NaOH)

- * Consider the titration of acid mixture (HCl & CH_3COOH) with strong base, NaOH

- * On addition of NaOH , the conductance of the acid mixture solution decreases steeply. The conductance decreases steeply because the small & highly mobile H^+ ions in the strong acid is gradually replaced by heavier & less mobile Na^+ ions.



- * Addition of base beyond the first neutralization point leads to moderate increase in conductance.

- * The conductance increases moderately, because the poorly conducting weak acid (CH_3COOH) is gradually converted to highly conducting salt [sodium acetate] until the second neutralization.



Further addition of NaOH, increases number of highly mobile OH^- ions. Hence conductance increases steeply.

